

Kindergarteners' peer relationships and early school absenteeism

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ABSTRACT

This study used a short-term longitudinal design to examine the extent to which kindergarteners': (a) peer relationships are associated with their school absenteeism; and (b) development of positive peer relationships vary as a function of their time spent in school. To address these aims, data were drawn from 801 kindergarteners across 64 classrooms and 15 schools in a Midwest city in the United States. Results from covariate-adjusted regression models revealed that kindergarteners' relational bonds were not predictive of their time away from school, and that their relationships with classmates were generally unaffected by absenteeism. There was, however, some indication that children's relationships with their classmates were more strongly predicted by their absenteeism for children from less educated households relative to children from more educated households.

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Every year, millions of American kindergartners miss excessive time from school, with one in four missing at least 10% of the school year (Chang and Davis, 2015; Romero and Lee, 2007). These high rates of absenteeism are of great concern because children's school success, even in kindergarten, is predicated on them showing up (Ansari and Gottfried, 2021; Ansari and Purtell, 2018; Connolly and Olson, 2012; Gershenson, Jacknowitz, and Brannegan, 2017; Morrissey, Hutchison, and Winsler, 2014). Given the prevalence and educational consequences of missing school, there has been growing interest among researchers, policymakers, and educators in better understanding the non-academic outcomes of absenteeism and the reasons why children are absent, especially in the earliest years of school. To date, however, much of the dialogue surrounding the causes of absenteeism has been on characteristics of children themselves and the role of their families (e.g., Gottfried and Gee, 2017) with little attention paid to other aspects of children's lives, including their relationships with their classmates. The above is a critical gap in knowledge because the relational bonds children share with their classmates are associated with their school adjustment and academic success (Buhs, 2005; Perdue, Manzeske, and Estell, 2009). Accordingly, from a policy and practice point of view, a richer portrait of the antecedents and outcomes of absenteeism taking

children's peer social experiences into account would provide schools and districts with opportunities to identify new solutions. Doing so would also allow for a better understanding of the wide-ranging consequences of missing school and, at the same time, point to opportunities for designing and implementing new intervention and prevention programs.

To address these gaps in knowledge, we shift the focus from the academic outcomes of absenteeism and the experiences of older children to children's peer social experiences at the onset of their educational careers. As part of this effort, we unpack the associations between different aspects of children's peer relationships and their absenteeism at the age of 5 in kindergarten. In doing so, we consider: (a) the extent to which peer relationships are associated with children's school absenteeism; and (b) the degree to which the cultivation of positive peer relationships is proportional to the time children spend in school. We also examine the extent to which the links between absenteeism and students' peer relationships vary for different groups of children that have been historically at risk for absenteeism in kindergarten, including children who experience socioeconomic disadvantages, children in poor health, and children without prior formal educational experiences (Gottfried and Hutt, 2019; Huang and Zhao, 2019; Ready, 2010). Taken together, our work can provide important insight into the ways in which efforts surrounding absenteeism and peer relationships can be tailored to different groups of children at the outset of their educational careers

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in order to maximize investments and promote more positive outcomes.

The antecedents of student absenteeism: A focus on early peer relationships

According to bioecological theory (Bronfenbrenner and Morris, 1998), children's development, including their school absences, is shaped by the interplay of their own characteristics, their environments, and the ways in which they engage with these environments. Drawing from this framework, previous studies of kindergartners and elementary school students have largely focused on child, family, community, and school characteristics that may contribute to school absences (Gottfried and Gee, 2017; Morrissey et al., 2014). To date, however, little attention has been paid to children's social experiences within classrooms, including their relationships with peers, which is a critical source of support for children to start and complete kindergarten successfully (Rubin, Bukowski, and Bowker, 2015). The above is a critical gap in applied knowledge because schools are increasingly interested in understanding what factors within school boundaries contribute to absenteeism (Kearney and Graczyk, 2022). In addition, several studies have shown that the degree to which children successfully navigate kindergarten is partially dependent on their relationships with peers, which has both short- and long-term influences on their social, emotional, and academic adjustment (Engle, McElwain, and Lasky, 2011; Ladd, Ettekal, and Kochenderfer-Ladd, 2017; Vitaro, Boivin, Brendgen, Girard, and Dionne, 2012). For example, positive peer relationships in the early and middle childhood years have been found to facilitate the formalization of self-concept and attitudes towards others (Nelson, Rubin, and Fox, 2005) as well as provide opportunities to learn and maintain social interaction with peers (Buhs, Ladd, and Herald, 2006). In contrast, negative peer relationships in early and middle childhood have been found to be associated with greater levels of externalizing behavior, conflict, and school avoidance (Criss, Pettit, Bates, Dodge, and Lapp, 2002), including in the kindergarten year (Buhs and Ladd, 2001).

When considering the role of peer relationships and student absenteeism, there are several different aspects of children's relationships with classmates that are of importance, including their number of friends, social preferences, receipt of social support, and experiences with victimization (Boele et al., 2019; Bukowski, Cillessen, and Velasquez, 2012; Howes, Rubin, Ross, and French, 1988; Ladd, Kochenderfer, and Coleman, 1997), which capture four conceptually independent aspects of classroom relationships. We discuss these constructs in more detail below. But before doing so, it is important to acknowledge that during the early years of school, children's reactions towards school are what is likely to influence their parents' decisions about whether to send them to school as children may insist on staying home, or alternatively, push to go to school (Kearney, 2007). For example, among children between the ages of 6 and 15, those who are emotionally distressed or those who want to avoid negative experience are particularly vulnerable to school refusal behavior (Havik, Bru, and Ertesvåg, 2015; Ingul, Klöckner, Silverman, and Nordahl, 2012; Lawrence, Dawson, Houghton, Goodsell, and Sawyer, 2019), which is seen as a way for children to persuade their parents into allowing them to stay home (Kearney and Albano, 2004). This type of school refusal behavior is a common behavior among kindergartners (e.g., Buhs and Ladd, 2001). Thus, although young children cannot decide whether to stay home on their own, there is evidence to support the notion that children who are emotionally distressed as a result of classroom experiences do find ways to stay home from school because they influence their parents' decisions.

With that above in mind, the number of friendships, which is bilateral and represents voluntary, intimate, and reciprocal relationships with peers (Howes et al., 1988), and social preferences, which capture the likeability relative to dislikeability of the group towards an individual child (Bukowski et al., 2012), have been found to be critical features of the relational environment. The extant literature has shown

that both the number of friendships and social preferences serve as a secure base for children to engage in the classroom environment and are associated with reductions in feelings of loneliness, social dissatisfaction, and engagement in school refusal behavior during the elementary school years (Antonopoulou, Chaidemenou, and Kouvava, 2019; Buhs, 2005; Buhs and Ladd, 2001; Ladd et al., 1997). In addition, children's friendships and social preference also matter for their access to positive play opportunities and partners, which may contribute to their enjoyment of school and likelihood of attending school. For example, studies have found that during the early years, friends engage in more frequent positive interactions, including playing in more complex and collaborative ways, offering more instrumental help, and managing conflict and disagreement better as compared with non-friend dyads (Tomada, Schneider, and Fonzi, 2002; van Hoogdalem, Singer, Eek, and Heesbeen, 2013). In contrast, young children with lower social preference scores have been found to spend more time alone and engage in less conversation and cooperative play with peers (Boulton, Don, and Boulton, 2011; Walker, 2009). That said, two prior studies which measured school avoidance in kindergarten with a combination of nurse visits and school absences found no association between children's positive peer relationships and their school avoidance (Ladd, 1990; Ladd and Price, 1987).

Peer social support and victimization may also associate with children's absenteeism in distinct ways. Peer social support reflects the receipt of social resources from peers that allow children to overcome difficulties and adjust to school (Ladd, Kochenderfer, and Coleman, 1996) and functions as a protective factor for children to better deal with anxiety and stress and cope with difficulties via guidance and feedback (Bakalim and Taşdelen Karçay, 2016). To the extent that social support helps children cope with difficulties, then it may help reduce children's tendency to be absent by reducing the frequency with which they engage in school refusal behavior. Additionally, the degree to which children perceive support from peers in the later elementary school years has been shown to be associated with children's school liking and engagement (Boulton et al., 2011; Estell and Perdue, 2013; Perdue et al., 2009). Victimization on the other hand, captures the extent to which a child is physically (e.g., hit, kicked, or shoved), verbally (e.g., teased), or relationally (e.g., excluded) attacked (Kochenderfer and Ladd, 1996). Victimization has been associated with the number of friends, social preferences, and receipt of social support, which tap into acceptance and friendship (Ladd et al., 1997). Due to the fear of attack or harm at school, early victims show increasing levels of loneliness and school avoidance and decreasing social satisfaction and school liking both at the end of the kindergarten and in the older grades (Kochenderfer and Ladd, 1996; Ladd et al., 2017). It is also likely for early victims to be isolated and less liked by peers, which leads to fewer playmates and playing opportunities (Godleski, Kamper, Ostrov, Hart, and Blakely-McClure, 2015).

The outcomes of absenteeism: A focus on the development of peer relationships

To understand the links between absenteeism and the development of students' relationships with peers, it is important to draw attention to the developmental and education literatures more generally. To that end, faucet theory contends that children's development (broadly defined) is proportional to their exposure to the classroom environment (Entwistle, Alexander, and Olson, 2001). Furthermore, because kindergarten is the primary setting for young children to engage in learning activities outside of their home (Justice, Jiang, Purtell, Lin, and Ansari, 2021), missing school has been argued to have a wide range of consequences. Reflecting these arguments, a series of quasi-experimental studies have shown that, above and beyond the risk factors associated with absenteeism, children who miss more instructional time during the early years demonstrate less optimal performance on assessments of math, science, and literacy (Ansari and Purtell, 2018; Connolly and Olson, 2012; Gershenson et al., 2017; Morrissey et al., 2014). These

studies also reveal that the links between absenteeism and students' testing performance in the early years of education are more pronounced for students at-risk (Ansari and Purtell, 2018; Ready, 2010).

Despite the growing evidence regarding the academic outcomes of absenteeism, far less attention has been paid to the non-academic outcomes of missing school. This gap in knowledge is surprising because not only do absent students receive fewer hours of instruction, which results in fewer opportunities to develop academic skills, but they also have fewer opportunities to interact with teachers and classmates, which is likely to result in less optimal social development (Ansari and Pianta, 2019; Finn, 1989). In kindergarten, children spend approximately 40% of their time in the classroom in group and child-selected activities, which involves interaction, collaboration, and negotiation with peers (Ansari and Purtell, 2017; Justice et al., 2021). Thus, children who miss school have fewer opportunities to establish and maintain relationships with peers and learn social-behavioral skills (e.g., identify and engage in positive communication skills, identify common conflicts and describe possible causes), which are identified as critical for kindergartners' social and emotional learning (Eklund, Kilpatrick, Kilgus, and Haider, 2018; Rimm-Kaufman et al., 2002). Reflecting these arguments, the few studies that have examined the relations between absenteeism and children's social outcomes in middle childhood and adolescence have found that absenteeism was linked to more behavioral problems and engagement in risky behaviors both in the short- and long-term (Ansari and Pianta, 2019; Finn, 1989; Hallfors et al., 2002). One study using a national sample of kindergartners also found that chronic absentees (i.e., those who missed 10% or more of classroom instructional time) demonstrated declines in their classroom social and behavioral engagement (Gottfried, 2014).

Thus, although peer relationships are likely to be predictive of children's school attendance, it is important to acknowledge that the development of peer relationships is also likely to be conditional on children showing up to school (Entwisle et al., 2001). On the one hand, one of critical conditions for relationship development is propinquity, which refers to the geographic and spatial distance between individuals (Newcomb, 1956). Attending school allows children to share the same space with classmates and creates the physical proximity for children to develop relationships. With that said, peer interactions in the classroom can be both positive and negative (Godleski et al., 2015). During the early years of education, children who are the target of verbal, relational, or physical aggression, attending school are more likely to be victimized in the absence of intervention (Vlachou, Andreou, Botsoglou, and Didaskalou, 2011). On the other hand, missing school limits the opportunities for children to practice interpersonal skills. Reflecting these possibilities, studies have shown that school absenteeism throughout children's educational careers is associated with elevated levels of disengagement and internalizing symptoms, less optimal social interactions, and more behavioral problems (Ansari and Pianta, 2019; Gottfried, 2014), which in turn, have been shown to be associated with less optimal peer relationships (Bandon, Calkins, Grimm, Keane, and O'Brien, 2010; Godleski et al., 2015). Along these same lines, school avoidance in the first months of kindergarten has been found to be associated with fewer friendships at the end of the year (Ladd, 1990).

Variability in the links between peer relationships and student absenteeism

The extant literature has made clear that kindergartners are at greater risk for absenteeism when they are from socioeconomically disadvantaged families (Ready, 2010), in poor health (Gottfried and Gee, 2017), and have no prior formal educational experiences (Gottfried, 2015). Just as importantly, students in early childhood (e.g., Shehu, 2019) and adolescence (e.g., Huang and Zhao, 2019) who experience such risks are also more likely to be rejected by peers. Beyond these main effects of vulnerability on absenteeism and peer relationship formation, there is growing evidence to suggest that kindergartners who come from more disadvantaged backgrounds (e.g., those from lower-

income and less educated homes) are more subject to the negative effects of absenteeism due to the limited resources available to them to compensate for what they missed when absent (Ready, 2010). In a similar vein, the association between lower quality relationships and children's development in middle childhood has been found to be more pronounced for children who come from families with fewer resources to support their learning (Criss, Shaw, Moilanen, Hitchings, and Ingoldsby, 2009). Thus, children from under resourced homes may be more susceptible to the effects of peer relationships and school absenteeism.

On the other hand, early educational experiences have been found to serve as a protective factor for children at risk as it equips them with the skills needed to succeed in kindergarten and beyond (Phillips et al., 2017). Early educational experiences are also known to result in improvements in parents home-school connections during the kindergarten year (Ansari and Crosnoe, 2015). As such, families whose children have prior educational experiences may be better able to leverage past experiences in order to navigate poor peer relationships and school absences. It is also well known that even though children miss school for many reasons in kindergarten, poor child health is a key correlate of school absenteeism (Gottfried and Gee, 2017). Children who miss time from school because of poor health miss school for different reasons than children in good health. Thus, when considering the antecedents and outcomes of school absenteeism, it is especially important to consider whether the associations between peer relationships and absenteeism vary as a function of children's overall health status.

Despite this accumulating body of evidence, and the different plausible patterns of moderation outlined above, whether these child and family risk factors obscure the links between children's relationships with peers and their absenteeism (and vice versa) remains understudied. Thus, it is critical to determine whether the links between absenteeism and peer relationships are similar for all students or whether some students are more affected than others. As part of the current investigation, we do so by examining differential associations by household income, maternal education, child health status, and prior educational experiences.

The current study

When taken together, the literature reviewed above suggests associations between different aspects of peer relationships and a wide range of school adjustment indicators across children's educational careers (e.g., Buhs and Ladd, 2001; Estell and Perdue, 2013; Hutzell and Payne, 2018; Ladd et al., 1997). For example, Ladd et al., 1997 demonstrated that friendship, peer acceptance, and victimization were associated with a set of school adjustment outcomes including expressions of loneliness, school liking, and school avoidance. Although the above school adjustment outcomes are correlated with school absenteeism and are likely a key mechanism for the links between peer relationships and absenteeism (e.g., Ingul and Nordahl, 2013; Ingul et al., 2012; Valiente, Lemery-Chalfant, Swanson, and Reiser, 2008), few studies have explored whether incoming peer relationships predict early school absenteeism (for exceptions see: Ladd, 1990; Ladd and Price, 1987). The above is of note because school absenteeism is viewed as the more critical indicator of children's long-term educational prospects and has implications for policy and practice given its inclusion in state accountability plans (Gottfried and Hutt, 2019). In addition, few studies have considered whether the development of peer relationships across the school year is predicated on children showing up to school. Thus, from an applied developmental perspective, the lack of focus on peer relationships and absenteeism has limited our understanding of the ways in which we can help improve children's educational experiences.

As such, the goal of the present investigation is to address gaps in knowledge regarding the relational predictors and outcomes of kindergarten absenteeism. Specifically, we address the following research questions: (1) To what degree do different aspects of kindergartners'

relationships predict their school absenteeism throughout the kindergarten year? (2) To what extent do kindergartners' school absenteeism throughout the school year associate with the changes in peer relationships over time? And (3) Do the links between kindergarten absenteeism and students' peer relationships vary as a function of risk status? Given the exploratory nature of the present study, and because few studies of absenteeism and children's peer relationships exist in the early years of school, we do not make directional hypotheses. But by addressing these research questions, the results of the present investigation have applied implications and the potential to facilitate a deeper understanding of the antecedents and outcomes of kindergarten absenteeism. For example, if findings suggest that absenteeism is associated with children's development of peer relationships, these findings can help educators and school districts use resources more strategically to help children navigate the classroom relational environment (Bierman, 2011). Likewise, using data to understand the extent of the absenteeism problem and why children are more likely to be absent is the first step in addressing the issue and ensuring that children show up to school on a regular basis. Thus, when taken together, findings from the current study can be incorporated in multi-tiered systems of support that seek to improve children's classroom experiences.

Method

Participants

Data were collected as part of the Early Learning Ohio study, a federally funded project aimed at understanding the classroom ecology from preschool to third grade (Justice et al., 2021). For the current study, we used data collected on five-year-olds during the kindergarten year in 2017–2018 due to the multiple assessments of children's peer relationships in the fall and spring. The study sample consisted of 801 kindergartners ($M_{\text{age}} = 67.24$ months, $SD = 4.45$) across 64 classrooms and 15 schools, representing almost 47% of the district's kindergartners. Almost half of the children were girls. Sixty eight percent of children were White, 9% were Black; 3% were Asian; 11% were Multi-racial, and 8% reported Other. Sixteen percent of children were Hispanic. A small number of children spoke a language other than English at home (15%) and had an Individualized Education Plan on file (6%). More than a third of children came from families that earned less than \$30,000; the average family income fell in the \$50,001–60,000 category. Eighty six percent of mothers had completed a high school degree or greater. With regards to children's teachers, all had attained at least a bachelor's degree (Bachelor's degree, 27%; Master's degree, 73%) and the average class size was 25.49 children ($SD = 0.86$). For other sample demographic information, see Table 1.

Procedures

This study received approval by the Institutional Review Board at The Ohio State University (2016B0231). Teachers and children were recruited from a suburban district in a midwestern city. A small portion of the sample (19%) was recruited in the year before kindergarten at age 4 and followed through kindergarten at age 5. In the fall of kindergarten, an additional group of children was added to meet recruitment goals. Before the school year started, teachers located within the district were invited to attend informational sessions, and those who consented to participate were enrolled in the study. Consenting teachers were asked to provide information on their classrooms, their students, and their own background via surveys. All children in classrooms taught by participating teachers were eligible to enroll. Consent packets were sent to parents via backpack mail and background information on children and their families was obtained through questionnaires as part of the active consent process. The average active classroom consent rate was 68% and ranged from 19% to 88%. In both the fall and spring of kindergarten, trained project staff interviewed and tested actively consented children

Table 1
Descriptives of study covariates.

	Mean (SD)/ Percent	Min	Max	Skewness	Kurtosis
<i>Child and family characteristics</i>					
Child age	67.24 (4.45)	52	92	0.53	4.18
Child gender: male	51.44	–	–	–	–
Child race		–	–	–	–
White	68.45	–	–	–	–
Black	8.8	–	–	–	–
Asian	3.19	–	–	–	–
Multi-racial	11.11	–	–	–	–
Other	8.43	–	–	–	–
Child ethnicity:					
Hispanic	15.70				
IEP status	6.33	–	–	–	–
Child primary language is English	87.84	–	–	–	–
Early educational experience at age 4					
Non attender	27.85				
Part-time attender (<20 h per week)	35.19				
Full-time attender (20+ hours per week)	36.96				
Household size	4.51 (1.32)	2	9	0.56	3.78
Number of children under 18	2.48 (1.15)	1	9	1.12	5.72
Household income	5.05 (4.40)	0	20	1.12	4.03
Maternal education					
Less than high school degree	14.00				
High school education with diploma	45.18				
Associate's degree	16.91				
Bachelor's degree	16.94				
Graduate or professional degree	7.27				
Marital status					
Married	61.58				
Living with partner/registered domestic partnership	11.86				
Separated/divorced/widowed	11.00				
Never married	15.46				
<i>Teacher and classroom characteristics</i>					
Highest degree of education (%)					
Bachelor's degree	26.83				
Master's degree	73.17				
Teaching experience (years)	12.62 (8.15)	2	35	0.68	3.03
Class size	25.49 (0.86)	23	27	–1.12	3.43
<i>Racial/ethnic composition (%)</i>					
White	63.72 (31.28)	0.00	100.00	–0.69	2.56
Black	12.75 (21.11)	0.00	100.00	2.67	10.55
Hispanic	17.24 (21.72)	0.00	100.00	1.97	7.63
Multiracial	10.99 (15.55)	0.00	100.00	3.21	17.93
Asian/Other	12.53 (18.35)	0.00	100.00	2.83	13.12
% of boys in classroom	52.13 (5.85)	0.00	64.00	0.08	2.25
% of children with IEP/IFSP	7.95 (5.66)	0.00	23.08	0.67	3.17
% of children with limited English	13.62 (17.90)	0.00	80.00	1.49	4.98
		0.00	95.00	–0.52	1.96

(continued on next page)

Table 1 (continued)

	Mean (SD)/ Percent	Min	Max	Skewness	Kurtosis
% of children receiving free/ reduced price lunch	55.55 (31.20)				
Classroom quality	4.25 (0.65)	2.86	5.54	0.15	2.29

during the school day in a quiet location. For a subgroup of children who participated only in the peer nomination aspect of data collection, passive consent was used, which on average represents 15% of children in the classroom (range = 0–73%).

Measures

Kindergarten absenteeism. Teachers reported the total number of days the target child was absent across the school year. Given that teachers completed the survey between April and May, we calculated the total number of absent days for each child based on a 180-day school year (see also, Ansari and Purtell, 2018). To do so, we estimated the number of days each child would have been absent given their absence rate and the remaining school days combined with teacher-reported days of absences. We also created a binary variable of chronic absenteeism (0 = child missed < 10% of the year; 1 = child missed 10% or more of the year; Balfanz and Byrnes, 2012).

Peer relationships. The number of mutual friendships and social preferences were measured through peer nominations in the fall and spring of kindergarten (Chen et al., 2020; Parkhurst and Asher, 1992). Trained interviewers asked the focal child to nominate an unlimited number of children in their class who fit a given description, including information about their best friends and who they liked to play with the most and least. To facilitate children's nomination, a photo roster of all children in the classroom was presented to them. The number of mutual friendships was counted for each focal child only when the child and classmate mutually nominated each other as best friends. Social preference was calculated according to the procedure devised by Coie and Dodge (1983); specifically, we subtracted the number of nominations a child received from classmates in terms of who they like the least from who they like the most. This approach to measuring peer social preference has been widely used in previous studies of young children (Ladd, 1990; Proulx and Poulin, 2013), and has demonstrated satisfactory psychometric properties (Babcock, Marks, Crick, and Cillessen, 2014; van den Berg and Cillessen, 2013).

Children were also interviewed in the fall and spring of kindergarten about their own experiences of peer victimization and social support in class. Four items in the self-reports of Peer Victimization Scale asked children to report the frequency with which they experienced four forms of peer victimization. More specifically, children reported the frequency with which their peers: (1) picked on them (general); (2) hit them (direct-physical); (3) said mean things about them (direct-verbal); and (4) said bad things about them (indirect-verbal; Kochenderfer and Ladd, 1996). Similarly, children responded to eleven items adapted from the Perceptions of Peer Social Support scale regarding the frequency they experienced support behaviors from peers (e.g., how often would kids in your class: Choose you on their team; helped you if you are hurt; Ladd et al., 1996; Waters, Lester, Wenden, and Cross, 2012). All items were rated on a three-point scale (0 = *Never*, 1 = *Sometimes*, 2 = *A lot*). Cronbach's alphas of the two scales ranged from 0.71 to 0.79. Composited scores for child-perceived peer victimization and peer social support at each time point was calculated for each child by averaging the responses from items in each scale separately.

Moderators. We focused on household income, maternal education, child health, and early educational experiences at age 4 as potential moderators that may differentiate the associations between absenteeism

and peer relationships. Information on these variables was gathered through parent surveys collected in the fall of the kindergarten year and again in the spring for those who did not respond to these questions in the fall. We compiled answers from the fall and spring parent survey to create indicators for each of our moderators. The annual household income included 21 categories with the lowest category ($\leq 10,000$) being coded as 0 and the highest category ($> 200,000$) being coded as 20. Maternal education was captured with five categories, including less than high school, high school education with diploma, associate's degree, bachelor's degree, and graduate or professional degree. Child health was rated on a 5-point scale ranging from *poor* (0) to *excellent* (4). For early educational experiences at age 4, parents were first asked if their child was enrolled in any center- or school-based program at age 4 and then asked about the number of hours of attendance per week. Based on their reports, we grouped children into nonattenders, part-time attenders (< 20 h per week), and full-time attenders (20+ hours per week; see also: Ansari, 2018; Bassok, Gibbs, and Latham, 2019). According to Rhemtulla, Brosseau-Liard, and Savalei (2012), variables containing five categories or more can be treated as continuous variables, and as such, household income, maternal education, and child health were included as continuous variables in the following analysis.

Covariates. To reduce the possibility of spurious associations, we controlled for a rich set of covariates that are theoretically and empirically informed to correlate with children's school absenteeism and peer experiences (e.g., Gottfried and Gee, 2017; Lin, Justice, Paul, and Mashburn, 2016; Poulin and Chan, 2010; Rubin et al., 2015). First, we controlled for *child characteristics*, including age, gender, race and ethnicity, IEP status, whether English is the primary home spoken language, parent report of health status, and early educational experiences at age 4. The second set of covariates were *household characteristics*, including household size, the number of children under 18, household income, maternal education, and mothers' marital status (married, living with partner, separated/divorced/widowed, never married). The last category captured the *characteristics of teachers and classrooms* the child attended, including teachers' highest degree of education, teaching experiences, class size, classroom quality as measured by the Classroom Assessment Scoring System (Pianta, La Paro, & Hamre, 2008), and classroom composition (race/ethnicity, gender, IEP status, limited English language skills, and free/reduced-price lunch receipt).

Variables included in sensitivity checks. In one of our sensitivity checks which we outline in more detail below, we include children's peer social skills in the fall and spring of kindergarten, which were measured using eight items from the Teacher-Child Rating Scale (T-CRS; Hightower et al., 1986). These items contained statements describing either positive (4 items; e.g., makes friends easily) or negative peer social behaviors (4 items; e.g., disturbs others while they are working). Teachers were asked to rate the degree to which they agreed with each statement according to their observation of each child. All items were rated on a 5-point Likert scale (0 = *strongly disagree*, 4 = *strongly agree*) and demonstrated high internal consistency ($\alpha = 0.95$ for both time points). After reverse coding the four negative items, we averaged teacher's ratings on all items for each child at each time point and used the mean score in the sensitivity checks.

Analytic approach

All analyses were performed in Mplus 7.4 (Muthén and Muthén, 2015). Missing data for study variables ranged from 1 to 27%, with family demographic information having the highest rate of missingness. Little's MCAR test suggested that data were not missing completely at random ($\chi^2(4556) = 6111.10, p < .001$). With that said, further examination of the missing data revealed that they were associated with child, teacher, and classroom characteristics. Based on the many patterns of missingness along with the correlations between key covariates and missing data, we assumed that data were likely to be missing at random (MAR; Schafer, 2002). Given the high possibility of MAR, Full

Information Maximum Likelihood (FIML) was used to address missing data. By adjusting the likelihood function and using all available data from each case to estimate the value of observed variables, FIML produces less biased parameter estimates and preserves analytic power compared with listwise/pairwise deletion (Enders, 2013). In addition, we clustered standard errors at the classroom level with Maximum Likelihood Robust estimation to account for the nesting of children within classrooms and to address potential issues of nonnormality (McNeish, Stapleton, and Silverman, 2017).

Following the general framework outlined above, we employed a variety of regression models according to the distribution of the outcome variables. Because two outcomes, i.e., days absent and number of mutual friendships, are count data and there are potential issues of overdispersion, we estimated Poisson and Negative Binomial regression models and compared these models for each outcome based on the likelihood ratio (LR) test. If the test is significant, the overdispersion is present and the negative binomial model with a freely estimated scaling parameter fits better (Coxe, West, and Aiken, 2009; Hung, 2012). The LR test statistic for days absent was significant ($\chi^2(df) = 28,122.08-26,916.26 = 1205.82(1), p < .001$), which suggests that the negative binomial regression *does* improve model fit compared to Poisson regression. On the other hand, the LR test statistic for number of mutual friendships was not significant ($\chi^2(df) = 23,452.95-23,453.18 = -0.23(1), p > .05$), which suggests that the negative binomial regression does *not* improve model fit compared to Poisson regression. Therefore, the negative binomial regression was used when models considered the number of days children were absent as the outcome whereas a Poisson regression was used when the outcome was the number of mutual friendships.

To test the extent to which children's peer relationships in the fall contributed to their frequency of absenteeism and likelihood of chronic absenteeism in kindergarten, we estimated negative binomial regression and linear probability models, respectively. After establishing the associations between peer relationships in the fall and children's absenteeism, we then regressed each of the peer relationships in the spring on days absent and chronic absenteeism, separately, with the corresponding fall scores and covariates controlled. As noted above, Poisson regression was selected to estimate the number of mutual friendships in the spring; the rest of peer relationships variables were estimated within an Ordinary Least Squares regression framework. Finally, to test whether the links between kindergartners' absenteeism and their peer relationships varied for different subgroups of children, interaction terms were added between the focal predictor(s) of interest and family income, maternal education, children's health status, and early educational experiences. Note that for all regression models estimated within an Ordinary Least Squares framework, we standardized all the continuous predictors and outcomes to have a mean of 0 and standard deviation of 1. However, for models with count data as outcomes, we preserved the scale of variables given the nature of the model.

Results

We begin with a discussion of the descriptive patterns of kindergartners' absenteeism and their peer relationships. We then turn to our research questions regarding the links between absenteeism and children's relational experiences and close with a series of sensitivity checks.

Descriptive patterns of kindergartners' absenteeism and peer relationships

As shown in Table 2, children missed an average of 11 days of school ($SD = 9.81$) during the kindergarten year, which translates to approximately 6% of the school year. In addition, 14% of children were identified as being chronically absent. With that said, there was a great deal of variability in the absenteeism status of the study children as indicated by the standard deviation and range (0–93). For instance, children who

Table 2
Descriptives of key variables.

	Mean (SD)/ Proportion	Min	Max	Skew	Kurtosis
Days absent	11.23 (9.81)	0	93	3.39	22.69
Percent chronically absent	14.09				
Fall peer relationships					
Social preferences	0.70 (2.95)	−14	10	−0.55	5.42
Mutual friendships	0.78 (0.96)	0	5	1.27	4.44
Child reported peer social support	1.29 (0.45)	0	2	−0.50	2.78
Child reported victimization	0.60 (0.59)	0	2	0.81	2.64
Spring peer relationships spring					
Social preferences	0.37 (2.90)	−13	9	0.64	4.93
Mutual friendships	1.01 (1.06)	0	6	1.15	4.37
Child reported peer social support	1.31 (0.41)	0	2	−0.49	2.89
Child reported victimization	0.65 (0.57)	0	2	0.61	2.41

were chronically absent (i.e., those who missed 10% or more of classroom instructional time) missed 28.6 days of kindergarten, whereas their counterparts who were not chronically absent missed only 8.4 days.

With regards to children's peer relationships, all indicators were weakly to moderately correlated ($r = |0.00-0.51|$; see Table 3). In terms of number of friendships, children on average were gaining one mutual friend across the year ($M_{fall} = 0.78, M_{spring} = 1.01$); in contrast, peer social preferences decreased across the year from 0.70 to 0.37. Additionally, children perceived that they had some peer social support both in fall and spring ($M = 1.29$ and 1.31) and the average frequency of their reported victimization ranged between never and sometimes ($M = 0.60$ and 0.65).

The antecedents of student absenteeism. Having established the descriptive patterns of absenteeism and kindergartners' peer relationships, we next analyzed the focal associations of interest. As presented in the top panel of Table 4, results from these analyses suggested that children's peer relationships in the fall were *not* significantly associated with children's rates of absenteeism ($b = |0.00-0.11|$) nor chronic absenteeism ($b = |0.01-0.03|$) in kindergarten. One exception was that children's reported victimization in the fall was positively associated with the number of days children were absent across the school year [$b = 0.11, \exp(b) = 1.12, SE = 0.06, p < .05$]. Practically speaking, a one unit increase in children's reported victimization was associated with a 12% increase in their rates of absenteeism. Overall, however, what these results imply is that kindergartners' incoming relationships are *not* an antecedent of school absenteeism.

Despite the largely null associations between peer relationships and student absenteeism, several of the child and family characteristics were associated with children's absenteeism. More specifically, children: Who had an IEP [$b = 0.23, \exp(b) = 1.25, SE = 0.12, p < .05$; $b = 0.17, SE = 0.07, p < .05$], whose mothers had never married [$b = 0.33, \exp(b) = 1.40, SE = 0.10, p < .01$; $b = 0.12, SE = 0.06, p < .05$], who did not have prior educational experiences [versus those with full-day experiences; $b = 0.17, \exp(b) = 1.19, SE = 0.08, p < .05$; $b = 0.08, SE = 0.05, p < .10$], and who were less healthy [$b = -0.14, \exp(b) = 0.87, SE = 0.05, p < .01$; $b = -0.06, SE = 0.02, p < .01$] were more likely to be absent and chronically absent.

The outcomes of absenteeism

The next part of our analyses considered the extent to which children's absenteeism and chronic absenteeism contribute to their peer relationships in the spring after controlling for baseline peer relationships and other covariates. As shown in Table 5, results revealed

Table 3

Correlations among peer relationship variables in the fall (F) and spring (S).

	1	2	3	4	5	6	7	8	
1. Days of absence	–								
2. Peer social preferences (F)	–0.07	–							
3. Mutual friendships (F)	–0.06	0.37***	–						
4. Child reported peer social support (F)	0.05	0.03	0.07	–					
5. Child reported victimization (F)	0.08*	–0.12**	–0.06	0.15***	–				
6. Peer social preferences (S)	–0.09*	0.51***	0.28***	0.07	–0.09*	–			
7. Mutual friendships (S)	–0.05	0.27***	0.30***	0.05	0.01	0.36***	–		
8. Child reported peer social support (S)	0.08	0.13***	0.08	0.31***	0.04	0.11**	0.08*	–	
9. Child reported victimization (S)	0.02	–0.19***	–0.05	0.00	0.36***	–0.18***	–0.02	0.11**	–

Table 4

Models with days of absence and chronic absenteeism as outcomes and fall, spring, and averaged peer social experiences as predictors.

Predictor	Days absent			Chronic absenteeism	
	b	Exp(b)	SE	b	SE
<i>Peer social experiences in fall</i>					
Intercept	3.17	23.88	1.51	–	–
Mutual friendships	–0.02	0.98	0.03	0.01	0.02
Social preferences	0.00	1.00	0.01	–0.01	0.01
Child reported peer social support	–0.01	0.99	0.08	–0.01	0.04
Child reported victimization	0.11*	1.12	0.06	0.03	0.03
<i>Peer social experiences in spring</i>					
Intercept	3.17	23.68	1.52	–	–
Mutual friendships	0.00	1.00	0.03	0.00	0.01
Social preferences	–0.02	0.98	0.01	–0.01*	0.01
Child reported peer social support	0.11	1.12	0.08	0.04	0.04
Child reported victimization	–0.00	0.99	0.07	–0.02	0.03
<i>Average peer social experiences</i>					
Intercept	3.28	26.46	1.58	–	–
Mutual friendships	–0.03	0.97	0.04	0.01	0.02
Social preferences	–0.01	0.99	0.01	–0.01 ⁺	0.01
Child reported peer social support	0.11	1.12	0.09	0.02	0.04
Child reported victimization	0.09	1.10	0.07	0.01	0.04

Notes. *** $p < .001$. ** $p < .01$. * $p < .05$. + $p < .10$.**Table 5**

Models with peer social experiences in spring as outcomes and absenteeism as the focal predictor.

	b	Exp (b)	SE	b	Exp (b)	SE
<i>Outcome: Mutual friendship</i>						
Mutual friendship in fall	0.28***	1.32	0.04	0.28***	1.32	0.03
Days absent/Chronic absenteeism	–0.00	1.00	0.01	–0.07	0.96	0.13
<i>Outcome: Social preferences</i>						
Social preferences in fall	0.48**		0.03	0.47***		0.03
Days absent/Chronic absenteeism	–0.05		0.03	–0.50		0.33
<i>Outcome: Child reported peer social support</i>						
Child reported peer social support in fall	0.27***		0.03	0.27***		0.04
Days absent/Chronic absenteeism	0.00		0.00	0.06		0.05
<i>Outcome: Child reported victimization</i>						
Child reported victimization in fall	0.33***		0.04	0.33***		0.04
Days absent/Chronic absenteeism	–0.03		0.06	–0.04		0.08

Notes. *** $p < .001$. ** $p < .01$. * $p < .05$. + $p < .10$.

nonsignificant associations between kindergarteners' absenteeism and chronic absenteeism and their peer relationships, suggesting that children's development of peer relationships was *not* predicated on them showing up to school.

Variability in the links between absenteeism and peer relationships

Thus far, our analyses considered the average associations between absenteeism and students' peer relationships, but not the degree to which these associations vary for key subgroups of interest. Accordingly, our final set of analyses considered whether children's family income, maternal education, children's health status, and children's early educational experiences at age 4 moderate the focal associations of interest. Of the 40 interactions estimated, six were statistically significant at $p < .05$, which is more than two that we would expect by chance. With that said, four of these significant interactions were for maternal education as the moderator. Of these four interactions, however, three were with peer relationships as the outcomes. Thus, 75% of the estimated interactions between absenteeism and maternal education were statistically significant when predicting peer relationships (see Table 6). Probing these interactions revealed that children's relationships with peers as measured by mutual friendships, social preferences, and victimization were relatively stable across the absenteeism distribution among children from more educated households. In contrast, for children from households with fewer years of education, their peer relationships were more strongly influenced by their absenteeism (see Figs. 1–3). More specifically, frequent absentees from households with fewer years of education demonstrated fewer mutual friendships and lower social preference scores than less frequent absentees from households with fewer years of education. On the other hand, although the associations for victimization also suggested that the “effects” of absenteeism were more pronounced for children from less educated households, peer victimization was found to be highest at low levels of absenteeism and lowest at high levels of absenteeism. In all, however, the links between absenteeism and kindergarteners' peer relationships were far more similar than different across different subgroups of at-risk children.

Sensitivity checks

Our primary analytic specification considered whether children's incoming peer relationships were associated with their absenteeism across the school year. Doing so allowed us to infer the potential influence of peer relationships on children's absenteeism based on their temporal sequence. However, it is plausible that what matters more than students' incoming peer relationships is their relationships at the end of the year or their development of relationships across the year. Because the above are all plausible analytic choices, and to ensure that our findings are robust to these decisions (see Silberzahn et al., 2018), we estimated a series of sensitivity checks with spring peer relationships, as well as averaged peer relationships across fall and spring as predictors, separately (see the middle and bottom panel of Table 4). As before, most aspects of spring and averaged peer relationships were *not* significant

Table 6

Differences in the associations between peer relationships and absenteeism by group.

	<i>b</i>	<i>SE</i>
<i>Days absent</i>		
Mutual friendships * health status	0.03	0.05
Social preference * health status	0.01	0.02
Child reported peer social support * health status	0.03	0.11
Child reported victimization * health status	0.15	0.09
Mutual friendships * early educational experiences	0.03 ^a /0.13 ^{a,b}	0.08 ^a / 0.06 ^b
Social preference * early educational experiences	-0.02 ^a / 0.03 ^b	0.03 ^a / 0.03 ^b
Child reported peer social support * early educational experiences	-0.04 ^a / 0.21 ^b	0.19 ^a / 0.19 ^b
Child reported victimization * early educational experiences	-0.19 ^a / -0.04 ^b	0.12 ^a / 0.13 ^b
Mutual friendships * maternal education	0.03	0.03
Social preference * maternal education	0.02**	0.01
Child reported peer social support * maternal education	0.12 ⁺	0.07
Child reported victimization * maternal education	0.06	0.07
Mutual friendships * household income	0.01	0.01
Social preference * household income	0.00	0.00
Child reported peer social support * household income	-0.00	0.02
Child reported victimization * household income	-0.02	0.01
<i>Mutual friendships</i>		
Days absent * health status	0.00	0.01
Days absent * household income	0.00	0.00
Days absent * maternal education	0.01*	0.00
Days absent * part-time attenders vs. nonattenders	-0.01	0.01
Days absent * full-time attenders vs. nonattenders	-0.00	0.01
<i>Social preferences</i>		
Days absent * health status	0.02	0.04
Days absent * household income	-0.01	0.03
Days absent * maternal education	0.04*	0.02
Days absent * part-time attenders vs. nonattenders	0.02	0.07
Days absent * full-time attenders vs. nonattenders	0.01	0.08
<i>Social support</i>		
Days absent * health status	-0.00	0.04
Days absent * household income	0.01	0.03
Days absent * maternal education	0.04 ⁺	0.03
Days absent * part-time attenders vs. nonattenders	-0.21 ⁺	0.11
Days absent * full-time attenders vs. nonattenders	-0.20	0.13
<i>Victimization</i>		
Days absent * health status	-0.04	0.05
Days absent * household income	0.09*	0.04
Days absent * maternal education	0.06*	0.03
Days absent * part-time attenders vs. nonattenders	0.12	0.12
Days absent * full-time attenders vs. nonattenders	-0.05	0.15

Note. *** $p < .001$. ** $p < .01$. * $p < .05$. + $p < .10$.^a The coefficients and SEs correspond to the comparison between nonattenders and part-time attenders (those who attended early education for <20 h per week).^b The coefficients and SEs correspond to the comparison between nonattenders and full-time attenders (those who attended early education for >20 h per week).

predictors of children's absenteeism.

Next, our primary models did not include teacher ratings of children's peer social skills as a control variable because it is unclear whether children's social skills are a confounder of the focal associations of interest or mechanism through which peer relationships influence

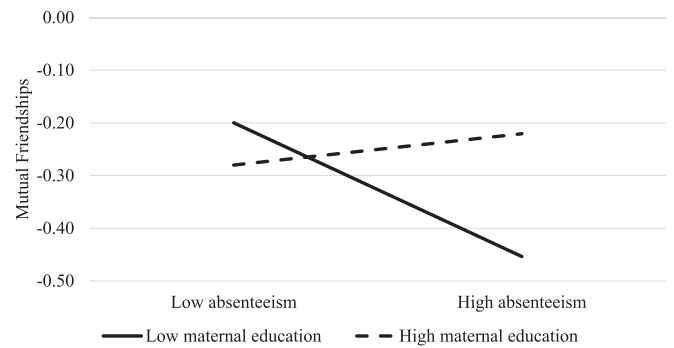


Fig. 1. An illustration of the conditional effects of absenteeism on children's mutual friendships as a function of maternal education. Note. Low and high levels of the predictor and moderator correspond to $-/+$ 1 standard deviation. Mutual friendships was estimated within a Poisson regression framework. The unstandardized coefficients of the intercept, predictor, moderator, and interaction term instead of exponentiated coefficients were used in this plot.

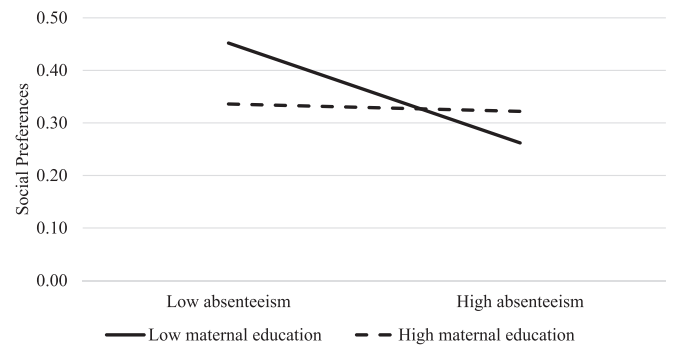


Fig. 2. An illustration of the conditional effects of absenteeism on children's social preferences as a function of maternal education. Note. Low and high levels of the predictor and moderators correspond to $-/+$ 1 standard deviation.

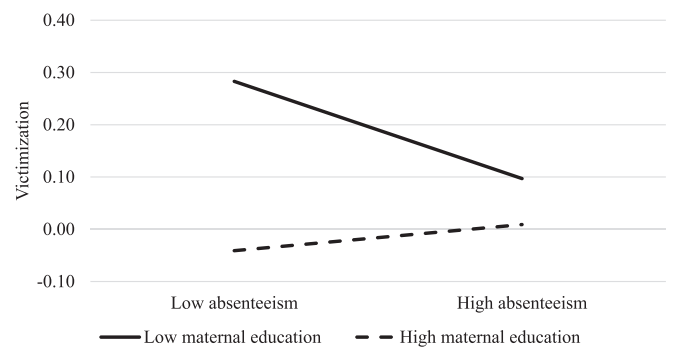


Fig. 3. An illustration of the conditional effects of absenteeism on children's victimization as a function of maternal education. Note. Low and high levels of the predictor and moderators correspond to $-/+$ 1 standard deviation.

student absenteeism. If children's social skills are a mechanism, then its inclusion would bias statistical inference (see Rohrer, 2018). But because it remains plausible that children's social skills is a confounder, we test additional models that consider whether the associations between peer relationships and days of absences/chronic absenteeism were sensitive to the inclusion of teacher-rated peer social skills. Results from these analyses revealed that teacher-rated peer social skills in the spring and across the school year were negatively associated with the number of days children were absent ($b = -0.15$, $SE = 0.05$, $p < .01$; $b = -0.14$, $SE = 0.05$, $p < .01$). But after controlling for teacher-rated peer social skills, all findings were similar to those reported above, suggesting

that children's social skills were not a confounding variable (nor mechanism) through which peer relationships influence absenteeism. One exception was that when we controlled for teacher-rated peer social skills in the fall, the association between children's reported victimization and their rates of absenteeism shrunk by 10% and became marginally significant [$b = 0.10$, $\exp(b) = 1.11$, $SE = 0.06$, $p < .07$].

Lastly, because eight children were absent >45 days across the school year, we conducted a sensitivity check in which we top coded these children at 45 and re-estimated all focal main effects models. When doing so, all findings were qualitatively and quantitatively similar as reported below (see supplemental Tables 1 and 2).

Discussion

Student absenteeism across K-12 has been of longstanding concern for educators and policymakers alike (Gottfried and Hutt, 2019). But among the growing body of studies that have considered the antecedents and outcomes of absenteeism (e.g., Gershenson et al., 2017; Morrissey et al., 2014), little attention has been paid to children's relational experiences inside the classroom setting. It is with this issue that the current investigation was primarily concerned. In addressing this issue, study results provided little support for the notion that absenteeism is either predicted by, or predictive of, the different aspects of kindergartners' relationships with their classmates. We discuss the results and implications of our work in more detail below.

To begin, the extant literature has made clear that a multitude of factors, including child, family, community, and school characteristics contribute to absenteeism in kindergarten (Ansari and Gottfried, 2018; Chang and Davis, 2015; Gottfried and Gee, 2017). And although prior studies reveal that the presence and likeability of friends serve as an emotionally secure base for young children to engage in the classroom environment (Antonopoulou et al., 2019; Buhs, 2005), we find that different aspects of children's relational bonds do *not*, on average, manifest in kindergartners' absenteeism nor rates of chronic absenteeism. Although somewhat surprising, these findings are in line with work by Ladd et al. (1987); Ladd (1990). One potential explanation for these null associations is that during the early years of education it is ultimately parents who make the decision whether their children show up to school (Kearney and Albano, 2004). Consequently, the null patterns reported herein may reflect parent's awareness (or lack thereof) of their children's classroom experiences. Another potential explanation is that children's peer relationships that support their school-going attitudes and school refusal behaviors, do not manifest in absenteeism because parents lack alternative care arrangements to accommodate their children not going to school. Thus, it is important that future studies consider: (a) other outcomes of children's relational experiences, such as their school refusal behavior that may in turn, have downstream effects on their school attendance; and (b) the diverse set of conditions under which children's relational experiences may manifest in absenteeism. If replicated in the future, what our results imply is that the quality and quantity of peer relationships are, on average, not critical for children's patterns of kindergarten absences, above and beyond other key correlates. Thus, schools and districts interested in improving school absenteeism are likely better off using their resources to address other factors that contribute to school attendance problems.

Next, our study adds to the ongoing discussions regarding the non-academic consequences of absenteeism. In forecasting the potential outcomes of absenteeism, we found that children's relational bonds were, for the most part, *not* proportional to their exposure to the classroom environment (Entwistle et al., 2001). Taken together, the findings reported herein stand in contrast with the extant literature that has shown clear and consistent associations between absenteeism in kindergarten and the early elementary school years and children's academic success (e.g., Ansari and Gottfried, 2021; Gershenson et al., 2017; Gottfried, 2014; Morrissey et al., 2014) and certain aspects of their social-behavioral development (e.g., Ansari and Gottfried, 2021;

Gottfried, 2014). In contrast to the extant literature, however, the present study focused on a unique aspect of children's classroom experiences that has received little attention. Despite these differences in outcomes, the general lack of associations may reflect the instability of peer relationships in the early years of schooling (Jiang and Cillessen, 2005; Monks, Smith, and Swettenham, 2003; Poulin and Chan, 2010). Indeed, as shown in the current study, there were low to moderate correlations between fall and spring peer relationships indicators ($r = 0.11$ – 0.51). Given the instability of peer relationships at this age, future studies should measure peer relationships at more than two time points in a year. But if our findings are replicated in the future, what they imply for practice is that school absenteeism does not pose a challenge to children's development and maintenance of friendships. Interventions and prevention programs that target absenteeism in kindergarten are therefore likely to yield little benefit for students' peer relational experiences.

Lastly, as part of the present study, we considered whether the associations between absenteeism and peer relationships are similar for children historically at risk for absenteeism and school difficulty and for those not at risk (Gottfried and Hutt, 2019). When considering the general pattern of results, we found that, for the most part, the links between students' absenteeism and their peer relationships did *not* vary as a function of children's socioeconomic status, health, nor prior educational experiences in the year before kindergarten. Put another way, the findings reported herein were not unique or more pronounced to children from specific backgrounds or with specific experiences. Instead, study findings were far more generalizable to kindergartners of different backgrounds. At the same time, however, there was some indication that household education moderated our focal findings of interest. For children from higher educated households, their school attendance did *not* matter for their development of peer relationships. In contrast, the children of parents with fewer years of education were at greater vulnerability to the effects of absenteeism. More specifically, children from households with fewer years of education were more negatively affected by absenteeism when considering their social preferences and mutual friendships as outcomes. And although absenteeism was also more strongly associated with victimization among children from less educated households, the pattern of vulnerability was different: Victimization was *highest* among children who were *infrequently* absent and from *less* educated households. When taken together, these different patterns of moderation imply that attending school is particularly influential for children from less educated households as it creates important opportunities for them to develop relationships, but these opportunities can result in both positive *and* negative outcomes (Godleski et al., 2015). One potential explanation for this pattern of moderation is that parents with more years of education are better able to help children navigate school absences and how to build relationships with peers and intervene when their children have negative peer experiences in school (Shehu, 2019), hence why no differences emerged among children from more educated homes.

As with any study, the results of the current investigation should be interpreted considering several limitations. To begin, our study sample was restricted to the experiences of kindergartners in a public school district in the state of Ohio. Accordingly, although our findings provide some of the first evidence regarding the links between absenteeism and peer relationships, our findings require replication with different samples across different grade levels and communities before conclusions and generalizations are made. Second, even though our analytic models employed a rich set of control variables and lagged dependent variables, our work is nonetheless non-experimental, and as such, our findings should be interpreted with caution. With that said, the questions posed as part of the current study do not readily lend themselves to a large, randomized control trial and to establish a causal link, a correlation is necessary (see also: Gershoff, Sattler, and Ansari, 2018). Additionally, our findings replicated when considering different analytic specifications, which lends confidence to our general conclusions. Third, we used

teacher-reported number of absent days for each child, which is one of the most common methods of reporting student absenteeism (e.g., Gottfried, 2017; Simon, Nylund-Gibson, Gottfried, and Mireles-Rios, 2020). Future studies, however, should consider accessing administrative records of individual children's absenteeism and also include information about *why* children were absent. Such information could provide a richer portrait of the focal associations of interest.

With these limitations and future directions in mind, the present investigation responded to calls in the extant literature that researchers consider the non-academic outcomes of absenteeism along with the different ecologies and experiences that shape students' school attendance (Gottfried and Hutt, 2019). In responding to these calls, our results provide some of the first evidence regarding the links between children's relational experiences in kindergarten and their school absenteeism. From an applied developmental perspective, this study explores the earliest life stage in which early peer relationships and the phenomenon of school absenteeism can emerge and mutually influence each other. Such findings (or lack thereof) are of note because schools and districts are increasingly trying to understand school absenteeism problems from an ecological perspective, including as a function of children's classroom experiences (Kearney and Graczyk, 2022). When taken together, study findings suggest that the kindergarten year may not yet be an onset of the associations between young children's relational bonds and their time away from school. There was, however, some indication that absenteeism was more strongly associated with the peer relationships of kindergartners from less educated households.

CRedit authorship contribution statement

Arya Ansari: Conceptualization, Methodology, Writing – original draft, Writing – review & editing. **Qingqing Yang:** Conceptualization, Formal analysis, Methodology, Writing – original draft, Writing – review & editing. **Kelly M. Purtell:** Conceptualization, Funding acquisition, Project administration, Writing – original draft. **Tzu-Jung Lin:** Conceptualization, Funding acquisition, Project administration, Writing – review & editing. **Laura M. Justice:** Conceptualization, Funding acquisition, Project administration, Writing – review & editing.

Data availability

Per Institute of Education Sciences policy, data will be made publicly available two years after project completion.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.appdev.2023.101529>.

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